F78PA Probability & Statistics A

Tutorial 10 – Solutions

1. Denoting by X the number of Heads in 20 flips, then $X \sim \text{Bin}(20, 0.2)$.

   (a) Markov's inequality: for a non-negative random variable X and a constant $\varepsilon > 0$, then $\mathbb{P}(X \geq \varepsilon) \leq \frac{\mathbb{E}(X)}{\varepsilon}$.
   Here we have $\mathbb{E}(X) = 20 \times 0.2 = 4$, so taking $\varepsilon = 16$, Markov's inequality gives

   $$\mathbb{P}(X \geq 16) \quad \leq \quad \frac{4}{16} = \frac{1}{4} = 0.25.$$

   (b) Chebyshev's inequality: for a random variable X with $\mathbb{E}(X) = \mu$ and $\mathbb{V}\text{ar}(X) < \infty$, and a constant $\varepsilon > 0$, then $\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\mathbb{V}\text{ar}(X)}{\varepsilon^2 1}$.
   Here we have $\mu = \mathbb{E}(X) = 4$ and $\mathbb{V}\text{ar}(X) = 20 \times 0.2 \times 0.8 = 3.2$. Then

   $$\begin{aligned} \mathbb{P}(X \geq 16) &= \mathbb{P}(X - \mu \geq 16 - 4) \\ &= \mathbb{P}(|X - \mu| \geq 12) \\ &\leq \frac{\mathbb{V}\text{ar}(X)}{12^2} \qquad \text{(Chebyshev)} \\ &= \frac{3.2}{144} = \frac{1}{45} \approx 0.0222. \end{aligned}$$

   [Note: in general, for $\varepsilon > 0$,

   $$\begin{aligned} \mathbb{P}(|X - \mu| \geq \varepsilon) &= \mathbb{P}(X - \mu \geq \varepsilon \text{ or } X - \mu \leq -\varepsilon) \\ &= \mathbb{P}(X - \mu \geq \varepsilon) + \mathbb{P}(X - \mu \leq -\varepsilon) \qquad \text{(disjoint events)} \\ &\geq \mathbb{P}(X - \mu \geq \varepsilon) \end{aligned}$$

   But in this particular case, with $X \sim \text{Bin}(20, 0.2)$, we have $\mathbb{P}(X - \mu \leq -\varepsilon) = \mathbb{P}(X - 4 \leq -16) = \mathbb{P}(X \leq -12) = 0$. Hence we obtain the equality $\mathbb{P}(X - \mu \geq 12) = \mathbb{P}(|X - \mu| \geq 12)$.]

   (c) The actual probability is

   $$\mathbb{P}(X \geq 16) \quad = \quad \sum_{k=16}^{20} \binom{20}{k} 0.2^k 0.8^{20-k} \approx 1.38 \times 10^{-8}.$$

   Comment: Chebyshev's inequality provides a substantially better bound than Markov's inequality, but still a bound that is far from the true probability.

2. Denote by X the height of a randomly sampled raccoon, in inches. Then $\mathbb{E}(X) = 10$ and has standard deviation $\sigma_X = 2$.

   (a) Using Chebyshev's inequality,

   $$\begin{aligned} \mathbb{P}(5 < X < 15) &= \mathbb{P}(|X - 10| < 5) \\ &= 1 - \mathbb{P}(|X - 10| \geq 5) \\ &\geq 1 - \frac{\mathbb{V}\text{ar}(X)}{5^2} \qquad \text{(Chebyshev)} \\ &= 1 - \frac{2^2}{5^2} = \frac{21}{25} = 0.84. \end{aligned}$$

(b) Now assume that X is normally distributed, so $X \sim \mathrm{N}\left(10, 2^2\right)$. Then

$$\begin{aligned}
\mathbb{P}(5 < X < 15) &= \mathbb{P}\left(\frac{5-10}{2} < \frac{X-10}{2} < \frac{15-10}{2}\right) \\
&= \mathbb{P}(-2.5 < Z < 2.5) \text{ where } Z \sim \mathrm{N}(0,1) \\
&= \Phi(2.5) - (1 - \Phi(2.5)) \\
&= 0.99379 - (1 - 0.99379) \qquad \text{(from tables)} \\
&= 0.98758.
\end{aligned}$$

The difference between the bound and the true probability is around 0.148, so bound is not very close to true answer.

3. Denote by X_i the length of the ith jump (in metres). Then the X_i are independent and identically distributed, with $X_i \sim \mathrm{Exp}(1)$, since an exponential random variable with mean 1 has rate parameter equal to 1.

$$\begin{aligned}
&\mathbb{P}(\text{Student must jump at least 60 times to finish the 50m race}) \\
&= \mathbb{P}\left(X_1 + X_2 + \cdots + X_{59} < 50\right)
\end{aligned}$$

Since $X_i \sim \mathrm{Exp}(1)$, we know that $E(X_i) = 1$ and $\mathbb{V}\mathrm{ar}(X_i) = 1$.

Writing $S = \sum_{i=1}^{59} X_i$, then $\mathbb{E}(S) = 59 \times \mathbb{E}(X_1) = 59$ and $\mathbb{V}\mathrm{ar}(S) = 59 \times \mathbb{V}\mathrm{ar}(X_1) = 59$.

By the Central Limit Theorem, since S is the sum of 59 i.i.d. random varialbes and 59 is large, we know that S is approximately normally distributed.

So

$$\begin{aligned}
&\mathbb{P}(\text{Student must jump at least 60 times to finish the 50m race}) \\
&= \mathbb{P}\left(X_1 + X_2 + \cdots + X_{59} < 50\right) \\
&= \mathbb{P}\left(\frac{S-59}{\sqrt{59}} < \frac{50-59}{\sqrt{59}}\right) \\
&\approx \mathbb{P}(Z < -1.17) \text{ where } Z \sim \mathrm{N}(0,1) \\
&= 1 - \Phi(1.17) \\
&= 1 - 0.8790 \qquad \text{(from tables)} \\
&= 0.121.
\end{aligned}$$

4. Denote by X the number of emails that take Anna more than one minute to answer. Then $X \sim \mathrm{Bin}(100, 0.6)$, under the assumption that the times taken to answer emails are independent of one another. So $\mathbb{E}(X) = 100 \times 0.6 = 60$ and $\mathbb{V}\mathrm{ar}(X) = 100 \times 0.6 \times 0.4 = 24$.

Using the Normal approximation to the Binomial distribution,

$$\begin{aligned}
\mathbb{P}(X \geq 65) &= \mathbb{P}(X \geq 64.5) \qquad \text{(continuity correction)} \\
&= \mathbb{P}\left(\frac{X-60}{\sqrt{24}} \geq \frac{64.5-60}{\sqrt{24}}\right) \\
&\approx \mathbb{P}(Z \geq 0.92) \text{ where } Z \sim \mathrm{N}(0,1) \\
&= 1 - \Phi(0.92) \\
&= 1 - 0.8212 \qquad \text{(from tables)} \\
&= 0.1788.
\end{aligned}$$

Note that we apply a continuity correction because X is a discrete random variable.

5. Denote by T_i the time (in hours) taken to mark script i for $i = 1, 2, \ldots, 100$.

   We are told that $\mathbb{E}(T_i) = 0.3$ and $\mathbb{V}\text{ar}(T_i) = 0.05$.

   Assumptions: We need to assume that $T_1, T_2, \ldots, T_{100}$ are identically distributed and independent.

   (a) For the initial stack of 27 scripts, let $S_{27} = T_1 + T_2 + \cdots + T_{27}$ be the total time taken to mark these scripts.
   Then $\mathbb{E}(S_{27}) = 27 \times \mathbb{E}(T_1) = 27 \times 0.3 = 8.1$ and $\mathbb{V}\text{ar}(S_{27}) = 27 \times \mathbb{V}\text{ar}(T_1) = 27 \times 0.05 = 1.35$. By the Central Limit Theorem, the distribution of S_{27} is approximately normal. So

$$\begin{aligned}
\mathbb{P}(S_{27} \leq 8) &= \mathbb{P}\left(\frac{S_{27} - 8.1}{\sqrt{1.35}} \leq \frac{8 - 8.1}{\sqrt{1.35}}\right) \\
&\approx \mathbb{P}(Z \leq -0.09) \text{ where } Z \sim \mathrm{N}(0, 1) \\
&= 1 - \Phi(0.09) \\
&= 1 - 0.5359 \qquad \text{(from tables)} \\
&= 0.4641.
\end{aligned}$$

   (b) For all 100 scripts, let $S_{100} = T_1 + T_2 + \cdots + T_{100}$ be the total time taken to mark these scripts.
   Then $\mathbb{E}(S_{100}) = 100 \times \mathbb{E}(T_1) = 100 \times 0.3 = 30$ and $\mathbb{V}\text{ar}(S_{100}) = 100 \times \mathbb{V}\text{ar}(T_1) = 100 \times 0.05 = 5$. By the Central Limit Theorem, the distribution of S_{100} is approximately normal.
   Denoting by x the number of hours that the lecturer allocates for marking all 100 scripts, we need to find x such that

$$\mathbb{P}(S_{100} \leq x) \geq 0.95.$$

   Now

$$\begin{aligned}
\mathbb{P}(S_{100} \leq x) &= \mathbb{P}\left(\frac{S_{100} - 30}{\sqrt{5}} \leq \frac{x - 30}{\sqrt{5}}\right) \\
&\approx \mathbb{P}\left(Z \leq \frac{x - 30}{\sqrt{5}}\right)) \text{ where } Z \sim \mathrm{N}(0, 1)
\end{aligned}$$

   From tables (Table 5), we have $\mathbb{P}(Z \leq 1.6449) = 0.95$, so we require

$$\begin{aligned}
\frac{x - 30}{\sqrt{5}} &\geq 1.6449 \\
\Rightarrow \qquad x &\geq 1.6449\sqrt{5} + 30 = 33.68
\end{aligned}$$

   The lecturer should allocate 33.68 hours for marking the 100 scripts.

6. (a) We are told that for $0 < x < 1$, the conditional density of Y given that $X = x$ is

$$f_{Y|X}(y \mid x) = \begin{cases} \frac{1}{x} & 0 < y < x, \\ 0 & \text{otherwise.} \end{cases}$$

   That is, for fixed $x \in (0, 1)$, the conditional density of Y is constant on the interval $(0, x)$. So the conditional distribution of Y given that $X = x$ is the Uniform distribution on the interval $(0, x)$.
   So $\mathbb{E}(Y \mid X = x) =$ expected value of $U(0, x)$ distribution $= x/2$.

(b) Law of Total Expectation:

$$\begin{aligned}\mathbb{E}(Y) &= \int_{x=0}^{1} \mathbb{E}(Y \mid X = x) f_X(x)\, dx \\ &= \int_{x=0}^{1} \left(\frac{x}{2}\right) 3x^2\, dx \\ &= \int_{x=0}^{1} \frac{3x^3}{2}\, dx = \left[\frac{3x^4}{8}\right]_{x=0}^{1} = \frac{3}{8} = 0.375\end{aligned}$$

(c) The region of non-zero density is $\{(x, y) : 0 < x < 1,\ 0 < y < x\}$, and so $\mathrm{Im}Y = (0, 1)$. Hence $F_Y(y) = 0$ for $y < 0$ and $F_Y(y) = 1$ for $y \geq 1$.
For $0 \leq y < 1$, by the Law of Total Probability:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \int_{x=0}^{1} \mathbb{P}(Y \leq y \mid X = x) f_X(x)\, dx$$

We are told that

$$f_X(x) = \begin{cases} 3x^2 & 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

We have seen in part (a) that the conditional distribution of Y given that $X = x$ is the Uniform distribution on the interval $(0, x)$, so

$$\mathbb{P}(Y \leq y \mid X = x) = F_{Y|X}(y \mid x) = \begin{cases} 0 & y < 0, \\ \frac{y}{x} & 0 \leq y < x, \\ 1 & y \geq x. \end{cases}$$

So for $0 \leq y < 1$,

$$\begin{aligned}F_Y(y) = \mathbb{P}(Y \leq y) &= \int_{x=0}^{1} \mathbb{P}(Y \leq y \mid X = x) f_X(x)\, dx \\ &= \int_{x=0}^{y} \mathbb{P}(Y \leq y \mid X = x)\ 3x^2\, dx + \int_{x=y}^{1} \mathbb{P}(Y \leq y \mid X = x)\ 3x^2\, dx \\ &= \int_{x=0}^{y} 1 \times 3x^2\, dx + \int_{x=y}^{1} \frac{y}{x} \times 3x^2\, dx \\ &= \int_{x=0}^{y} 3x^2\, dx + \int_{x=y}^{1} 3xy\, dx \\ &= \left[x^3\right]_{x=0}^{y} + \left[\frac{3x^2 y}{2}\right]_{x=y}^{1} = y^3 + \frac{3y}{2} - \frac{3y^3}{2} = \frac{3y}{2} - \frac{y^3}{2}\end{aligned}$$

So now

$$F_Y(y) = \begin{cases} 0 & y < 0, \\ \frac{3y}{2} - \frac{y^3}{2} & 0 \leq y < 1, \\ 1 & y \geq 1. \end{cases}$$

Marginal density for Y is then

$$f_Y(y) = \frac{dF_Y}{dy} = \begin{cases} \frac{3}{2} - \frac{3y^2}{2} & 0 \leq y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

7. Denote by X_i the time (in weeks) that the ith pen lasts. We are told that $E(X_i) = 1$ and $\mathbb{V}\mathrm{ar}(X_i) = 1/4$, and it's reasonable to assume that the lifetimes of pens are identically distributed and independent of one another.

If Sam buys n pens at the start of 30 weeks at university, then

$$\mathbb{P}(\text{Sam does not run out of pens}) \;=\; \mathbb{P}(X_1 + X_2 + \cdots + X_n \geq 30)$$

Write $S_n = X_1 + X_2 + \cdots + X_n$. We are asked to find the smallest n such that

$$\mathbb{P}(S_n \geq 30) \;\geq\; 0.95.$$

Now $\mathbb{E}(S_n) = n \times \mathbb{E}(X_1) = n$, $\mathbb{V}\text{ar}(S_n) = n \times \mathbb{V}\text{ar}(X_1) = n/4$, and the Central Limit Theorem says that provided n is large, the distribution of S_n is approximately normal.

So

$$\begin{aligned}
\mathbb{P}(S_n \geq 30) &= \mathbb{P}\left(\frac{S_n - n}{\sqrt{n/4}} \geq \frac{30-n}{\sqrt{n/4}}\right) \\
&\approx \mathbb{P}\left(Z \geq \frac{30-n}{\sqrt{n/4}}\right) \text{ where } Z \sim \mathrm{N}(0,1)
\end{aligned}$$

so we need to find n such that

$$\mathbb{P}\left(Z \geq \frac{30-n}{\sqrt{n/4}}\right) \;\geq\; 0.95.$$

From tables (Table 5), we have $\mathbb{P}(Z \geq 1.6449) = 0.05$, so by symmetry of the distribution, $\mathbb{P}(Z \geq -1.6449) = 0.95$.

So we need to find n such that

$$\frac{30-n}{\sqrt{n/4}} \;\leq\; -1.6449$$

To do this, set $x = \sqrt{n}$, and find x such that

$$\begin{aligned}
\frac{30-x^2}{x/2} &= -1.6449 \\
\Rightarrow \quad 30 - x^2 &= -0.82245x \\
\Rightarrow \quad x^2 - 0.82245x - 30 &= 0 \\
\Rightarrow \quad x &= \frac{0.82245 \pm \sqrt{0.82245^2 + 4 \times 1 \times 30}}{2} \qquad \text{(quadratic formula)} \\
&= -5.0814 \text{ or } 5.9039
\end{aligned}$$

Since $x = \sqrt{n}$, we know that x must be positive, so $x = 5.9039$, and we require $n \geq 5.9039^2 = 34.86$. So Sam should buy 35 pens.

8. X is the number of errors in a transmission of 2000 digits, and we are told that each digit is incorrectly transmitted with probability 0.001, independently of all other digits. That is, $X \sim \text{Bin}(2000, 0.001)$. Hence $\mathbb{E}(X) = 2000 \times 0.001 = 2$, and the distribution of X may be approximated by the Poi(2) distribution. So

$$\begin{aligned}
\mathbb{P}(X > 2) &= 1 - \mathbb{P}(X \leq 2) \\
&= 1 - (\mathbb{P}(X=0) + \mathbb{P}(X=1) + \mathbb{P}(X=2)) \\
&\approx 1 - (\mathbb{P}(Y=0) + \mathbb{P}(Y=1) + \mathbb{P}(Y=2)) \text{ where } Y \sim \text{Poi}(2) \\
&= 1 - \left(e^{-2} + \frac{2^1 e^{-2}}{1!} + \frac{2^2 e^{-2}}{2!}\right) \\
&= 1 - 5e^{-2} \approx 0.3233.
\end{aligned}$$

9. Told that $X \sim \text{Poi}(\lambda)$, $Y \sim \text{Poi}(\mu)$, and X, Y independent.

   Let $W = X + Y$. We have $\text{Im}X = \{0, 1, 2, \ldots\}$ and $\text{Im}Y = \{0, 1, 2, \ldots, \}$, and so $\text{Im}W = \{0, 1, 2, \ldots\}$.

   For $k = 0, 1, 2, \ldots,$

$$\begin{aligned}
\mathbb{P}(W = k) &= \sum_{i=0}^{k} \mathbb{P}\left(X = i,\ Y = k - i\right) \\
&= \sum_{i=0}^{k} \mathbb{P}(X = i)\mathbb{P}(Y = k - i) \qquad \text{since } X, Y \text{ independent} \\
&= \sum_{i=0}^{k} \left(\frac{\lambda^i e^{-\lambda}}{i!}\right)\left(\frac{\mu^{k-i} e^{-\mu}}{(k-i)!}\right) \\
&= \frac{e^{-(\lambda+\mu)}}{k!} \sum_{i=0}^{k} \frac{k!}{i!\,(k-i)!} \lambda^i \mu^{k-i} \\
&= \frac{e^{-(\lambda+\mu)}}{k!} \sum_{i=0}^{k} \binom{k}{i} \lambda^i \mu^{k-i} \\
&= \frac{e^{-(\lambda+\mu)}}{k!} (\lambda + \mu)^k \qquad \text{by the Binomial theorem}
\end{aligned}$$

   That is, the random variable W has the probability mass function of the $\text{Poi}(\lambda + \mu)$ distribution. So $W = X + Y \sim \text{Poi}(\lambda + \mu)$, as required.

Exam Style Question

(a) Suppose that $X_1, X_2, \ldots$ are independent and identically distributed random variables with $\mathbb{E}(X_i) = \mu$, $\mathbb{V}\text{ar}(X_i) = \sigma^2 < \infty$, and let $S_n = X_1 + \ldots X_n$ for $n = 1, 2, \ldots$
Then for *any* $a < b$,

$$\lim_{n\to\infty} \mathbb{P}\left(a \leqslant \frac{S_n - n\mu}{\sqrt{n\sigma^2}} \leqslant b\right) = \mathbb{P}\left(a \leqslant Z \leqslant b\right) \quad (*) \qquad \text{where } Z \sim \text{N}\left(0, 1\right)$$

(1 mark)

(b) We need to make the assumptions that $X_1, X_2, \ldots, X_{80}$ are independent and identically distributed random variables. (1 marks)
Under the independence assumption, then $\mathbb{E}[S] = 80 \times 2 = 160$ and $\mathbb{V}\text{ar}[S] = 80 \times 2 = 160$. (2 marks)
By the Central Limit Theorem, the distribution of S is approximately $\text{N}\left(160, 160\right)$. (1 mark)

The probability that all emails are answered by 11am is

$$\begin{aligned}
\mathbb{P}(S \leq 180) &= \mathbb{P}\left(\frac{S - 160}{\sqrt{160}} \leq \frac{180 - 160}{\sqrt{160}}\right) \\
&\approx \mathbb{P}(Z \leq 1.58) \qquad \text{where } Z \sim \text{N}\left(0, 1\right) \\
&= \Phi(1.58) \approx 0.9429. \qquad \text{(2 marks)}
\end{aligned}$$

Recall here Φ denotes the cumulative distribution function of the standard normal random variable and we calculated its value using the statistical tables.

(c) We require the value x such that

$$\mathbb{P}(S \leq x) = 0.99$$

This is equivalent to requiring

$$\mathbb{P}\left(\frac{S-160}{\sqrt{160}} \leq \frac{x-160}{\sqrt{160}}\right) = 0.99.$$

(2 marks)

From the statistical tables, for $Z \sim \mathrm{N}(0,1)$, we have $\mathbb{P}(Z \leq 2.3263) \approx 0.99$. (1 mark). So by the Central Limit Theorem

$$\mathbb{P}\left(\frac{S-160}{\sqrt{160}} \leq 2.3263\right) \approx 0.99.$$

Therefore we require (approximately)

$$\frac{x-160}{\sqrt{160}} = 2.3263$$

and hence

$$x = 160 + 2.3263\sqrt{160} = 189.43 \text{ minutes.}$$

(2 marks)